CAFFEINE & CADAVER
MBBS BIOCHEMISTRY NOTES SERIES

Biological Oxidation

MBBS BIOCHEMISTRY · PAPER 1

Redox Potential | ETC Complexes I-IV | Chemiosmotic Theory | ATP Synthase | ETC Inhibitors | Uncouplers | Malate Shuttle | G3P Shuttle | Brown Adipose Tissue | Xenobiotics | Free Radicals | Antioxidants

🧠 📊 EXAM PRIORITY MAP & USAGE STRATEGY

- ✦ 🔴 **TIER 1 (Must Do):** High yield, frequently tested concepts and PYQs. Core for passing.
- ✦ 🟡 **TIER 2 (Moderate):** Important mechanisms, good for scoring well.
- ✦ 🟢 **TIER 3 (Low Yield):** Extra details, mainly for Viva and distinction.

⏱ **Usage Guide: 6h = Tier 1; 3h = Tier 1 + Viva; 1h = Last Day Revision only.**

00 🔥 Last Day Revision Mode

ULTRA-CONCISE EXAM FACTS

✦ **ETC Protons:** Complex I ($4H^+$), III ($4H^+$), IV ($2H^+$). **Complex II pumps ZERO protons.**

✦ **ATP Yield:** 1 NADH = 2.5 ATP. 1 $FADH_2$ = 1.5 ATP. Total aerobic glucose = 32 ATP.

✦ **Inhibitors:** Rotenone (I), Malonate (II), Antimycin A (III), Cyanide/CO (IV), Oligomycin (V).

✦ **Uncouplers:** DNP & Thermogenin (BAT). Dissipate H^+ gradient as heat → ↑ O_2 consumed, ↓ ATP made.

✦ **Shuttles:** Malate-Aspartate = 2.5 ATP (Heart/Liver). Glycerophosphate = 1.5 ATP (Brain/Muscle).

✦ **Xenobiotics:** Phase I (CYP450 Oxidation) & Phase II (Glucuronidation Conjugation).

✦ **Free Radicals:** Hydroxyl (•OH) is most destructive. SOD converts $O_2^{\bullet -}$ → H_2O_2.

01 Redox Potential & Overview

TIER 3

▶ **Expand for Viva Details (Low Yield)**

Biological oxidation = removal of electrons (or H atoms) from metabolic substrates. Unlike combustion, energy released in a **controlled, stepwise** manner → captured as ATP. Final electron acceptor = **O_2** → forms H_2O.

STANDARD REDOX POTENTIAL (E°') PYQ 2012-S

✦ Measure of tendency of a compound to **donate electrons** (reducing agent)

✦ Measured in **Volts (V)** at pH 7, 25°C, 1M concentrations

✦ More **negative E°'** = stronger reducing agent = better electron donor

✦ More **positive E°'** = stronger oxidising agent = better electron acceptor

✦ Electrons flow from **low (negative)** → **high (positive) E°'**

✦ $\Delta G° = -nF\Delta E°'$ (n = electrons, F = Faraday constant 96,500 J/V·mol)

Key E°' Values (approximate):

COUPLE	E°' (V)
NAD^+/NADH	−0.32
FAD/$FADH_2$	−0.22
CoQ/$CoQH_2$	+0.04
Cyt c (Fe^{3+}/Fe^{2+})	+0.23
O_2/H_2O	+0.82

Total $\Delta E°'$ = +0.82 − (−0.32) = **+1.14 V** → highly spontaneous

MEMORY HOOK

"Electrons run downhill — from NADH (most negative) to O_2 (most positive) — like water flowing to the sea."

▶ MASTER OVERVIEW — FLOW OF REDUCING EQUIVALENTS

Metabolic Substrates (Glycolysis, TCA, β-Oxidation) Oxidation → **NADH + H^+** and **$FADH_2$** produced (reducing equivalents = electron carriers)

↓ Enter inner mitochondrial membrane

Electron Transport Chain (ETC) — Complexes I → II → CoQ → III → Cyt C → IV
Electrons passed sequentially. Energy released at 3 proton-pumping sites → H^+ pumped into intermembrane space

↓ Proton gradient (electrochemical) drives ATP synthase

Oxidative Phosphorylation (Complex V — ATP Synthase) H^+ flows back through F_0F_1-ATPase → **ADP + Pi** → **ATP**

↓ Electrons + H^+ + O_2 (Complex IV)

Terminal Oxidation **$4H^+ + 4e^- + O_2 \rightarrow 2H_2O$** (catalysed by Cytochrome c Oxidase)

02 ETC — The 4 Complexes

● TIER 1

All ETC complexes are embedded in the **inner mitochondrial membrane (IMM)**. The IMM is impermeable to H^+, ions, and most polar molecules — this is ESSENTIAL for the proton gradient. Location: Cristae of mitochondria.

COMPLEX I — NADH DEHYDROGENASE ⭐ PYQ 2014-S, 2016-S, 2023

COMPLEX I (NADH : CoQ Oxidoreductase)

- ✦ **Substrate:** NADH + H^+ (from TCA, β-oxidation, glycolysis via shuttle)
- ✦ **Electron carrier components:** FMN (flavin mononucleotide) + 8–9 Iron-Sulfur (Fe-S) clusters
- ✦ **Reaction:** $NADH + H^+ + CoQ \rightarrow NAD^+ + CoQH_2$
- ✦ **Protons pumped:** 4 H^+ per NADH (matrix → intermembrane space)
- ✦ **ATP yield:** Contributes ~2.5 ATP per NADH
- ✦ **Inhibitor:** `Rotenone` (rat/fish poison), `Amytal` (barbiturate), `Metformin` (partial inhibition)

COMPLEX II — SUCCINATE DEHYDROGENASE PYQ 2016-S

COMPLEX II (Succinate : CoQ Oxidoreductase)

- ✦ **Substrate:** $FADH_2$ (generated by succinate dehydrogenase in TCA — same enzyme!)
- ✦ **Electron carrier components:** FAD + Fe-S clusters + Cytochrome b_{560}
- ✦ **Reaction:** Succinate + FAD → Fumarate + $FADH_2$ → passes electrons to CoQ

✦ **Protons pumped: ZERO** — does NOT pump protons → explains lower ATP yield from $FADH_2$

✦ **ATP yield:** ~1.5 ATP per $FADH_2$

✦ **Inhibitor:** `Malonate` (competitive inhibitor of succinate dehydrogenase — classic biochemistry experiment)

> ⚠️ ☠️ **EXAM TRAP:** Complex II is the ONLY ETC complex that does NOT pump protons. This is why $FADH_2$ yields fewer ATP than NADH.

COMPLEX III — CYTOCHROME BC_1 COMPLEX ⭐ PYQ 2014-S, 2016-S, 2023

COMPLEX III ($CoQH_2$: Cytochrome c Oxidoreductase)

✦ **Substrate:** $CoQH_2$ (reduced ubiquinone from Complex I and II)

✦ **Components:** Cytochrome b (with bL and bH haems), Rieske Fe-S protein, Cytochrome c_1

✦ **Mechanism:** Q cycle — $CoQH_2$ donates $2e^-$ and $2H^+$; electrons transferred via Cyt b → Cyt c_1 → Cyt c (mobile carrier)

✦ **Protons pumped:** 4 H^+ per 2 electrons (Q cycle effectively doubles pumping)

✦ **Inhibitor:** `Antimycin A` (antibiotic), `BAL (British Anti-Lewisite)`

COMPLEX IV — CYTOCHROME C OXIDASE ⭐ PYQ MULTIPLE YEARS

COMPLEX IV (Cytochrome c Oxidase — Terminal Oxidase)

✦ **Substrate:** Reduced Cytochrome c (accepts 4 electrons one at a time)

✦ **Components:** Cu_A, Cytochrome a, Cytochrome a_3, Cu_B (binuclear centre a_3-CuB is O_2 binding site)

✦ **Reaction:** 4 Cyt c(Fe^{2+}) + $4H^+$ + O_2 → 4 Cyt c(Fe^{3+}) + $2H_2O$

✦ **Protons pumped:** 2 H^+ per 2 electrons (= 4 H^+ per O_2)

✦ **Inhibitor:** `Cyanide (CN-)`, `Azide`, `CO`, `H₂S` — all bind Fe^{3+} of Cyt a_3 → block terminal electron transfer → LETHAL

> CYANIDE POISONING — MECHANISM
>
> CN^- binds tightly to the Fe^{3+} of cytochrome a_3 (Complex IV) → prevents electron transfer to O_2 → ETC stops → ATP production stops → cell death. Tissues that die fastest: brain and heart (highest O_2 demand). **Treatment:** Hydroxocobalamin (binds CN^-) or Nitrites (generate metHb that competes with CN^-) + Thiosulphate (converts CN^- → thiocyanate).

ETC COMPLEXES — SUMMARY TABLE ⭐ EXAM FAVOURITE

COMPLEX	NAME	ELECTRON DONOR	ELECTRON ACCEPTOR	H^+ PUMPED	INHIBITOR
I	NADH Dehydrogenase	NADH	CoQ	4 H^+	Rotenone, Amytal
II	Succinate Dehydrogenase	$FADH_2$	CoQ	**0 (ZERO!)**	Malonate
III	Cytochrome bc_1	$CoQH_2$	Cyt c	4 H^+	Antimycin A
IV	Cytochrome c Oxidase	Cyt c	O_2	2 H^+	CN^-, CO, Azide
V	ATP Synthase (F_0F_1)	H^+ gradient	—	—	Oligomycin

03 Mobile Carriers — CoQ & Cytochrome C

TIER 3

▶ **Expand for Viva Details (Low Yield)**

Two mobile carriers connect the ETC complexes by diffusing within or along the inner mitochondrial membrane. **PYQ 2013-S asked specifically: "Ubiquinone and Cytochrome C have special role in respiratory chain as mobile carriers."**

COENZYME Q (UBIQUINONE / COQ)

✦ **Nature:** Lipid-soluble benzoquinone with isoprenoid tail (CoQ10 in humans — 10 isoprene units)

✦ **Location:** Freely mobile within the lipid bilayer of IMM

✦ **Function:** Collects electrons from both Complex I and Complex II → delivers to Complex III

✦ **Redox forms:** Oxidised (Q) → semiquinone ($Q^{\bullet-}$, radical) → fully reduced (QH_2 / ubiquinol)

✦ **Special role:** Acts as an antioxidant — can quench free radicals directly

✦ **CoQ10 as supplement:** Used in heart failure — replenishes CoQ10 which ↓ in aging and statin use

CYTOCHROME C

PYQ 2013-S, 2024

✦ **Nature:** Small peripheral membrane protein (12 kDa), loosely associated with outer face of IMM

✦ **Location:** Intermembrane space — mobile along membrane surface

✦ **Function:** Transfers electrons from Complex III → Complex IV (one electron at a time — Fe^{2+} ↔ Fe^{3+})

✦ **Special role in APOPTOSIS:** Release of Cyt c from mitochondria → cytoplasm → activates **Apaf-1** → **Caspase-9** → **Caspase-3 cascade** → programmed cell death. PYQ 2024

✦ **EQ trigger:** "Cytochrome C has important role in apoptosis" [2024]

04 Chemiosmotic Theory & ATP Synthase

TIER 1

Proposed by Peter Mitchell (1961) — Nobel Prize 1978. Key insight: Energy from electron transport is stored as a **proton electrochemical gradient** (proton motive force), not as a chemical intermediate.

⭐ PYQ 2010, 2023

MITCHELL'S CHEMIOSMOTIC HYPOTHESIS

⭐ PYQ 2010, 2023

▶ MECHANISM OF OXIDATIVE PHOSPHORYLATION

Step 1 — Electron Transport Complexes I, III, IV pump H^+ from **matrix → intermembrane space (IMS)**
Builds proton gradient: IMS is acidic (high $[H^+]$) + positive; Matrix is alkaline + negative

↓ Creates Proton Motive Force (PMF) = ΔpH + ΔΨ (membrane potential)

Step 2 — Proton Motive Force (PMF) Two components: **Chemical gradient (ΔpH)** = ~0.75 V + **Electrical gradient (ΔΨ)** = ~0.14 V
Total PMF ≈ **0.22 V** (or ~200 mV) across IMM

↓ H^+ can ONLY re-enter matrix through ATP Synthase (F_0F_1)

Step 3 — ATP Synthesis (F_0F_1 ATPase / Complex V) H^+ flows down gradient through F_0 (rotor in membrane) → rotates c-ring → mechanical energy transmitted to F_1 (in matrix) → conformational changes in β-subunits → **ADP + Pi → ATP**

KEY REQUIREMENT — INTACT MITOCHONDRIAL MEMBRANE

The IMM must be **intact and impermeable to H^+** for the gradient to be maintained. If the membrane is disrupted (detergents, mechanical damage) or if H^+ is allowed to leak back without going through ATP synthase → gradient dissipates → no ATP synthesis → energy released as heat. This is how **uncouplers** work.

F_0F_1 ATP SYNTHASE — STRUCTURE

PYQ 2019-S "F0F1 ATPASE"

F_0 (membrane sector):

- ✦ Embedded in IMM — forms the H^+ channel
- ✦ Subunits: a, b_2, c-ring (9–12 c subunits)
- ✦ H^+ flows through interface between a-subunit and c-ring → drives rotation
- ✦ **Oligomycin** blocks H^+ channel of F_0 → stops ATP synthesis

F_1 (matrix sector — catalytic):

- ✦ $\alpha_3\beta_3\gamma\delta\varepsilon$ arrangement
- ✦ β-subunits are catalytic — have 3 conformations: Open (O), Loose (L), Tight (T)
- ✦ Binding change mechanism: rotation of γ-subunit changes β conformation → releases ATP
- ✦ **~3 H^+ per ATP synthesised** (each rotation makes 3 ATP from 3 β-subunits)

ATP SYNTHASE ANALOGY

"F_0 is the water wheel (H^+ spins it), F_1 is the generator (converts spin → ATP). Mitchell's dam = the IMM."

05 ATP Yield Calculation ● TIER 1

⚡ Modern P/O ratios (NADH = 2.5 ATP, $FADH_2$ = 1.5 ATP) — based on H^+/ATP ratio of ATP synthase. Older textbooks used 3 and 2 — MBBS board exams may still accept these. Know both!

⭐ Frequently calculated

COMPLETE OXIDATION OF GLUCOSE — ATP TALLY

STAGE	NADH	$FADH_2$	GTP/ATP (DIRECT)	ATP FROM NADH (×2.5)	ATP FROM $FADH_2$ (×1.5)
Glycolysis (cytoplasm)	2 NADH	—	2 ATP	5* (via shuttle)	—
Pyruvate → Acetyl-CoA (×2)	2 NADH	—	—	5	—
TCA Cycle (×2 turns)	6 NADH	2 $FADH_2$	2 GTP	15	3
TOTAL	**10 NADH**	**2 $FADH_2$**	**4**	**25**	**3**

Total ATP = 25 + 3 + 4 = 32 ATP (modern calculation)

Older calculation: 10 NADH × 3 = 30 + 2 $FADH_2$ × 2 = 4 + 4 = **38 ATP** (still accepted in MBBS boards)

*Cytoplasmic NADH: if via Malate shuttle → 2.5 ATP; if via Glycerophosphate shuttle → 1.5 ATP

⚠️ ☠️ **EXAM TRAP:** Don't confuse the old (38 ATP) and new (32 ATP) yields. MBBS boards often accept 38, but 32 is biochemically accurate based on 2.5/1.5 P/O ratios.

06 ETC Inhibitors ● TIER 1

⚡ ETC inhibitors **BLOCK electron flow** → both electron transport AND ATP synthesis stop simultaneously. The proton gradient cannot form. This is different from uncouplers (which allow electron flow but uncouple it from ATP synthesis).

INHIBITOR	SITE (COMPLEX)	MECHANISM	CLINICAL/TOXICOLOGICAL RELEVANCE
Rotenone	Complex I	Blocks electron transfer from Fe-S to CoQ. Lipophilic — crosses membranes easily	Rat/fish poison; used in research. Linked to Parkinson's disease (mitochondrial toxin)
Amytal (Amobarbital)	Complex I	Same as rotenone — blocks Fe-S → CoQ step	Barbiturate sedative — high doses cause respiratory failure via ETC inhibition
Metformin	Complex I (mild)	Partial inhibition → ↓ NADH oxidation → ↑ AMPK → ↓ gluconeogenesis	First-line diabetes drug; rarely causes lactic acidosis via complex I inhibition
Malonate	Complex II	Competitive inhibitor of succinate dehydrogenase (structural analogue of succinate)	Classic competitive inhibition experiment (overcome by excess succinate)
Antimycin A	Complex III	Binds Qi site — blocks $CoQH_2$ reoxidation in Q cycle → stops Cyt b reduction	Fungicide / fish poison; important research tool
BAL (Dimercaprol)	Complex III	Chelates Fe-S clusters in Rieske protein	Originally developed as antidote for arsenical gas (Lewisite) in WWII; still used in heavy metal poisoning
Cyanide (CN^-) LETHAL	Complex IV	Binds Fe^{3+} of Cyt a_3 → blocks O_2 binding → terminal electron transfer stops	Industrial (electroplating), accidental, suicidal. Treatment: hydroxocobalamin + nitrites + thiosulphate
Carbon Monoxide (CO)	Complex IV	Binds Fe^{2+} of Cyt a_3 (also binds Hb). 250× affinity for Hb vs O_2	Silent killer — house fires, car exhaust. Treatment: 100% O_2 / hyperbaric O_2
Azide (N_3^-)	Complex IV	Same as cyanide — binds Cyt a_3	Laboratory use; toxic
Oligomycin Antibiotic	Complex V (F_0)	Blocks H^+ channel of F_0 → H^+ cannot re-enter → backpressure builds → ETC slows/stops. NOT an ETC inhibitor per se — inhibits the ATPase	NEW 2025 PYQ "Oligomycin can inhibit respiratory chain" [2025]. Indirect inhibition.

🧠 INHIBITOR SITES MNEMONIC

"Rotenone Ruins I, Malonate Murders II, Antimycin Attacks III, Cyanide Crushes IV, Oligomycin Obliterates V"

R-M-A-C-O → "Really Murderous Animals Chase Owners"

⚠️ ☠️ **EXAM TRAP:** Oligomycin and cyanide both inhibit the respiratory chain, but differently! Cyanide is a direct ETC inhibitor (Complex IV). Oligomycin inhibits ATP synthase (Complex V) and only indirectly stops ETC due to proton backpressure.

07 Uncouplers of Oxidative Phosphorylation

TIER 1

Uncouplers allow the ETC to continue working (electrons still flow, O_2 still consumed) but the proton gradient is dissipated as **HEAT** instead of making ATP. They act as H^+ carriers — transporting protons across the IMM bypassing ATP synthase.

PYQ 2019, 2022

HOW UNCOUPLERS WORK

▶ UNCOUPLING MECHANISM

Normal: H^+ pumped out → gradient builds → H^+ can ONLY return via F_0F_1 → ATP made

↓ Uncoupler added

Uncoupler (e.g., DNP) picks up H^+ from IMS (protonated, lipophilic) → diffuses through IMM → releases H^+ into matrix (deprotonated, moves back)
Net result: H^+ gradient DISSIPATED. ATP synthase starved of H^+. ETC continues but energy = HEAT

↓ Result

↑ O_2 consumption + ↑ Heat production + ↓ ATP synthesis + ↑ Metabolic rate (to compensate) + ↑ Body temperature

UNCOUPLER	MECHANISM	CLINICAL/PHYSIOLOGICAL RELEVANCE
2,4-Dinitrophenol (DNP) TOXIC	Lipophilic weak acid — carries H^+ across IMM as H-DNP; releases in matrix as DNP^-. Classic experimental uncoupler	PYQ 2022 Used illegally as weight-loss drug in 1930s. Causes hyperthermia, sweating, tachycardia. LETHAL — narrow toxic:therapeutic ratio. Banned
Thermogenin (UCP-1) PHYSIOLOGICAL	Uncoupling Protein-1 in brown adipose tissue — natural H^+ channel in IMM — regulated by norepinephrine and fatty acids	Non-shivering thermogenesis in newborns, hibernating animals. Activated by cold → ↑ fatty acid oxidation → heat. PYQ 2011, 2016-S
Long-chain Fatty Acids	Act as weak acid protonophores at high concentrations — carry H^+ across IMM	Physiological modulation of coupling efficiency in mitochondria-rich tissues
Valinomycin	Ionophore for K^+ — dissipates membrane potential ($\Delta\Psi$) rather than ΔpH	Research tool; not clinically relevant
FCCP (Carbonyl cyanide p-trifluoromethoxyphenylhydrazone)	Most potent experimental protonophore — completely uncouples at nM concentrations	Research tool for studying mitochondrial respiration (Seahorse assay)
High-dose Aspirin (Salicylate)	Weak acid — protonophore at high concentrations	Aspirin overdose → fever (hyperthermia) + uncoupling + metabolic acidosis. Partially explains fever in salicylate toxicity

🧠 INHIBITOR VS UNCOUPLER — THE KEY DIFFERENCE

> **"Inhibitor: blocks BOTH electron flow AND ATP synthesis (chain stops). Uncoupler: allows electron flow but STEALS the H^+ gradient → heat not ATP."**

Think: Inhibitor = traffic jam. Uncoupler = road exists but leads nowhere useful.

⚠️ ☠️ **EXAM TRAP:** Inhibitors stop both ETC and ATP synthesis. Uncouplers stop ATP synthesis but INCREASE ETC activity and O_2 consumption (energy released as heat).

08 Membrane Shuttles

TIER 2

⚡ The IMM is impermeable to NADH. Cytoplasmic NADH (from glycolysis) must transfer its electrons into the mitochondria via one of two shuttle systems.

⭐ PYQ 2010-S (Malate), 2014

MALATE-ASPARTATE SHUTTLE

⭐ PYQ 2010-S, 2014

▶ MALATE-ASPARTATE SHUTTLE

CYTOPLASM NADH + OAA → Malate (by MDH) + NAD^+ [NADH oxidised]

↓ Malate enters mitochondria via malate-α-KG antiporter

MITOCHONDRIAL MATRIX Malate → OAA (by MDH) + NADH [NADH regenerated in matrix]

↓ OAA → Aspartate (transamination) → exported → back to OAA in cytoplasm

Net: 1 cytoplasmic NADH = 1 mitochondrial NADH → **2.5 ATP**

Active in: **Heart, liver, kidney** (high metabolic activity tissues with abundant aspartate aminotransferase)

GLYCEROPHOSPHATE SHUTTLE (G3P SHUTTLE)

▶ GLYCEROL-3-PHOSPHATE SHUTTLE

CYTOPLASM NADH + DHAP → Glycerol-3-P (by cytoplasmic G3P dehydrogenase) + NAD^+

↓ Glycerol-3-P transfers electrons to FAD on inner mitochondrial membrane

INNER MITOCHONDRIAL MEMBRANE (outer face) Glycerol-3-P + FAD → DHAP + $FADH_2$ (mitochondrial G3P dehydrogenase)

↓ $FADH_2$ feeds electrons to CoQ → Complex III onwards

Net: 1 cytoplasmic NADH → $FADH_2$ → **1.5 ATP only**

Active in: **Brain, skeletal muscle, white adipose tissue** (quick but less efficient)

COMPARING THE TWO SHUTTLES

PYQ 2014

FEATURE	MALATE-ASPARTATE	GLYCEROPHOSPHATE
End product in mitochondria	NADH (mitochondrial)	$FADH_2$
ATP yield per cytoplasmic NADH	2.5 ATP	**1.5 ATP**
Reversibility	Reversible	Irreversible
Key enzyme	MDH + Aspartate aminotransferase	Glycerol-3-phosphate dehydrogenase
Tissues	Heart, liver, kidney	Brain, skeletal muscle
Complexity	More complex (2 enzymes each side)	Simpler

SHUTTLE MEMORY HOOK

"Malate shuttle = More ATP (2.5) — heart loves efficiency. Glycerophosphate = Less ATP (1.5) — brain accepts less because it's faster."

09 Brown Adipose Tissue & Thermogenin

TIER 1

Brown fat generates heat (thermogenesis) WITHOUT shivering via controlled uncoupling. Key protein: Thermogenin (UCP-1).

PYQ 2011, 2016-S "Brown fat = non-shivering thermogenesis in newborn"

BAT VS WHITE ADIPOSE TISSUE

PYQ 2011, 2016-S, 2022

Why brown fat is BROWN:

- ✦ Rich in mitochondria → iron in cytochromes → brown colour
- ✦ Multilocular fat droplets (multiple small → WAT has one large)
- ✦ Dense capillary network for heat dissipation
- ✦ High UCP-1 (Thermogenin) content in IMM

UCP-1 Activation Cascade:

- ✦ Cold exposure → Sympathetic NS → Norepinephrine released
- ✦ β_3-adrenergic receptor → Adenylate cyclase → ↑ cAMP → PKA → Lipase activated → TAG → FFA released
- ✦ FFA bind to UCP-1 → channel opens → H^+ floods back into matrix → heat generated
- ✦ FFA also = fuel for β-oxidation → sustains ETC

CLINICAL RELEVANCE OF BAT

Newborns: Critical for survival — cannot shiver (underdeveloped muscle). BAT around neck, between shoulder blades, around kidneys provides thermogenesis. Premature babies lack sufficient BAT → hypothermia risk.
Adults: BAT persists around clavicle, mediastinum — more active in cold-adapted individuals. Active area of research for obesity treatment. ↑ BAT activity = ↑ energy expenditure = possible anti-obesity target.

10 Xenobiotics & Cytochrome P450 · TIER 2

Xenobiotics = foreign compounds (drugs, pollutants, food additives) not produced by normal metabolism. Biotransformation makes them water-soluble for excretion. Primary site: **Liver smooth ER**. PYQ 2016-S, 2023

PHASES OF XENOBIOTIC METABOLISM · PYQ 2016-S, 2023, 2025

▶ BIOTRANSFORMATION — PHASE I AND PHASE II

PHASE I — Functionalisation Reactions Introduce or unmask a functional group (-OH, $-NH_2$, -SH, -COOH)
Reactions: Oxidation (most common — via CYP450), Reduction, Hydrolysis
Location: SER of hepatocytes. May activate or inactivate drugs. May produce toxic/reactive intermediates.

↓ Phase I product usually slightly more polar — then conjugated in Phase II

PHASE II — Conjugation Reactions (Biosynthetic) Add large polar group to Phase I product → highly water-soluble → urinary/biliary excretion
Reactions: Glucuronidation (UDP-glucuronate — most common), Sulphation, Glutathione conjugation, Acetylation, Methylation, Amino acid conjugation (glycine, glutamine)
All Phase II reactions increase molecular weight and water solubility.

↓ Water-soluble metabolite excreted in urine or bile

Phase III (transport): ABC transporters (P-glycoprotein, MRP) export conjugates out of hepatocytes → bile or blood → kidney → excretion

CYTOCHROME P450 (CYP) SYSTEM · PYQ 2013, 2014-S, 2019, 2025

Structure & Location:

✦ Haemoproteins — Fe in porphyrin ring absorbs at 450 nm when bound to CO (hence "P450")

✦ Location: SER of liver (also intestine, lung, adrenal cortex)

✦ ~57 CYP genes in humans; CYP1A2, CYP2C9, CYP2D6, CYP3A4 handle most drugs

✦ CYP3A4 = most abundant hepatic CYP (~30–40% of total)

Overall Reaction:

$RH + O_2 + NADPH + H^+ \rightarrow R\text{-}OH + H_2O + NADP^+$

Requires: **NADPH-CYP450 reductase** (FAD + FMN containing) as electron donor, and **O_2**

One O atom → hydroxylation of substrate (R-OH)
Other O atom → water

This is a **monooxygenase / mixed-function oxidase** reaction.

CYP450 — CLINICAL IMPORTANCE

Drug interactions: CYP inducers (rifampicin, carbamazepine, phenytoin, St John's Wort) → ↑ CYP expression → faster drug metabolism → ↓ drug levels → therapeutic failure. CYP inhibitors (fluconazole, erythromycin, grapefruit juice) → ↓ drug metabolism → ↑ drug levels → toxicity.
Steroid synthesis: CYP enzymes in adrenal cortex perform multiple hydroxylation steps (CYP11A1, CYP21A2, CYP11B1).
Carcinogen activation: Benzo[a]pyrene (tobacco smoke) → CYP1A1 → reactive epoxide → binds DNA → mutation → cancer.

11 Free Radicals & Antioxidants ● TIER 1

Free radicals = molecules with unpaired electron in outer orbital → highly reactive, seek to pair electron → chain reactions of damage. Major source in biology: **mitochondrial ETC leakage** (1–3% of electrons escape → superoxide $O_2^{\bullet -}$).

PYQ 2011-S, 2012-S, 2014, 2015-S

REACTIVE OXYGEN SPECIES (ROS) — GENERATION & DAMAGE PYQ 2019

▶ ROS CASCADE

Primary ROS — Superoxide ($O_2^{\bullet -}$) $O_2 + e^- \rightarrow O_2^{\bullet -}$ (via Complex I/III leakage, NADPH oxidase in phagocytes, xanthine oxidase)

↓ Superoxide Dismutase (SOD) — Cu-Zn in cytoplasm, Mn in mitochondria

H_2O_2 — Hydrogen Peroxide $2\ O_2^{\bullet -} + 2H^+ \rightarrow H_2O_2 + O_2$ (catalysed by SOD). H_2O_2 is not a radical but is reactive

↓ Fe^{2+} (Fenton reaction) or Cu^+

Hydroxyl Radical ($^{\bullet}OH$) — MOST REACTIVE $H_2O_2 + Fe^{2+} \rightarrow {}^{\bullet}OH + OH^- + Fe^{3+}$ (Fenton reaction). Cannot be detoxified by any enzyme — attacks DNA, proteins, lipids

Targets of ROS Damage:

✦ **DNA:** 8-oxoguanine, strand breaks → mutations, cancer

✦ **Lipids:** Lipid peroxidation — chain reaction destroying membrane PUFAs → Malondialdehyde (MDA) — marker of oxidative stress

Beneficial ROS Actions:

✦ **Phagocyte respiratory burst:** NADPH oxidase generates $O_2^{\bullet -}$ → kills bacteria (neutrophils, macrophages)

✦ **Signal transduction:** H_2O_2 acts as second messenger — activates NF-κB, AP-1

✦ **Apoptosis:** ROS can trigger programmed cell death → tumour suppression

✦ **Proteins:** Carbonylation, cross-linking, enzyme inactivation

✦ **Carbohydrates:** Glycoxidation → AGEs (advanced glycation end-products) in diabetes

✦ EQ trigger: "Free radicals may have some beneficial role" [2015-S]; "Macrophages show beneficial effects by generating free radicals" [2021]

ANTIOXIDANT DEFENCES `PYQ 2011-S, 2012-S, 2014`

ANTIOXIDANT	TYPE	MECHANISM	CLINICAL NOTE
Superoxide Dismutase (SOD)	Enzymatic	$O_2^{\bullet-} \rightarrow H_2O_2 + O_2$. Cu-Zn SOD (cytoplasm), Mn-SOD (mitochondria), EC-SOD (extracellular)	ALS linked to mutant Cu-Zn SOD. First line defence
Catalase	Enzymatic	$2\ H_2O_2 \rightarrow 2\ H_2O + O_2$. Located in peroxisomes	G6PD deficiency → ↓ NADPH → ↓ GSH → ↓ catalase support → RBC haemolysis
Glutathione Peroxidase (GPx)	Enzymatic	H_2O_2 + 2 GSH → 2 GSSG + 2 H_2O (uses selenium — selenocysteine in active site)	Requires NADPH to regenerate GSH (via glutathione reductase). Links to HMP shunt
Vitamin E (α-Tocopherol)	Non-enzymatic (lipid-soluble)	Scavenges lipid peroxyl radicals ($LOO^{\bullet}$) in cell membranes → chain-breaking antioxidant → Vit E radical regenerated by Vit C	Deficiency → haemolytic anaemia in newborns, spinocerebellar ataxia. Works in membranes
Vitamin C (Ascorbate)	Non-enzymatic (water-soluble)	Scavenges $O_2^{\bullet-}$, $^{\bullet}OH$, peroxyl radicals in aqueous phase. Regenerates Vit E from Vit E radical	Also cofactor for prolyl hydroxylase (collagen), dopamine hydroxylase, enhances iron absorption
Glutathione (GSH)	Non-enzymatic (thiol)	Tripeptide (γ-Glu-Cys-Gly). Cys thiol donates electrons. Reduced by glutathione reductase (NADPH-dependent)	Detoxification (Phase II, paracetamol toxicity), membrane protection. `PYQ 2016-S, 2022`
β-Carotene	Non-enzymatic	Quenches singlet oxygen ($^1O_2$) — excited form of O_2. Pro-vitamin A	High-dose supplements may increase cancer risk in smokers (paradox)
Uric Acid	Non-enzymatic	Scavenges peroxyl radicals, $^{\bullet}OH$, and complexes iron. Major antioxidant in human plasma	High uric acid = antioxidant paradox — also causes gout

12 Viva Q&A

TIER 1

ETC & ATP SYNTHESIS

How do inhibitors of ETC differ from uncouplers of oxidative phosphorylation? PYQ 2011, 2015-S

ETC Inhibitors block electron flow at specific complexes → NO electrons flow → NO proton gradient forms → NO ATP + NO O_2 consumed. Complete shutdown.

Examples: Rotenone (Complex I), Antimycin A (Complex III), Cyanide (Complex IV).

Uncouplers allow electron flow to continue (O_2 IS consumed) but dissipate the proton gradient as heat instead of making ATP. They are protonophores — carry H^+ across IMM bypassing F_0F_1.

Examples: DNP (2,4-dinitrophenol), Thermogenin/UCP-1 (physiological), Salicylate.

Key exam point: Uncoupler → ↑ O_2 consumption + ↑ heat + ↓ ATP. Inhibitor → ↓ O_2 consumption + ↓ heat + ↓ ATP.

Explain the chemiosmotic theory of oxidative phosphorylation. PYQ 2010, 2023

Peter Mitchell's Chemiosmotic Hypothesis (1961):

- Energy of electron transport is conserved as an **electrochemical proton gradient** (proton motive force, PMF) across the IMM
- ETC complexes I, III, IV pump H^+ from matrix → IMS → creates pH gradient (ΔpH) + electrical potential (ΔΨ)
- IMM is impermeable to H^+ → gradient cannot dissipate except through F_0F_1-ATPase
- H^+ flows back through F_0 → rotates c-ring → γ-subunit of F_1 rotates → β-subunit conformational changes (Open → Loose → Tight) → ATP released
- ~3 H^+ per ATP synthesised; ~10 H^+ pumped per NADH oxidised

Evidence: (1) Isolated mitoplasts can make ATP if artificial pH gradient applied. (2) Oligomycin blocks H^+ channel → blocks ATP synthesis but also slows ETC (backpressure). (3) Membrane vesicles with bacteriorhodopsin + ATP synthase make ATP on light exposure — pure H^+ gradient → ATP.

What is the role of Cytochrome C in ETC and in apoptosis? `PYQ 2013-S, 2024`

Role in ETC: Mobile electron carrier — transfers electrons from Complex III (Cyt bc_1) to Complex IV (Cyt c oxidase). Small peripheral protein (12 kDa) loosely associated with outer face of IMM in the IMS. Transfers one electron at a time via Fe^{2+}/Fe^{3+} redox cycle.

Role in Apoptosis (Intrinsic/Mitochondrial pathway):

- Pro-apoptotic signals (DNA damage, oxidative stress) → Bax/Bak oligomerise in outer mitochondrial membrane → MOMP (mitochondrial outer membrane permeabilisation)
- Cyt c released into cytoplasm
- Cytoplasmic Cyt c + Apaf-1 + dATP → Apoptosome (heptameric wheel)
- Apoptosome recruits and activates Procaspase-9 → active Caspase-9 → activates effector Caspase-3 → cell dismantled

Thus Cyt c is a dual-function protein: life (ATP production) and death (apoptosis).

BROWN FAT, SHUTTLES & XENOBIOTICS

How does brown adipose tissue generate heat? What is thermogenin? PYQ 2011, 2016-S

Brown adipose tissue (BAT) generates heat by **non-shivering thermogenesis** through controlled uncoupling of oxidative phosphorylation.

Thermogenin (UCP-1 — Uncoupling Protein 1):

- A proton channel/carrier protein in the IMM of BAT mitochondria
- Normally closed — activated by: (a) Free fatty acids released during lipolysis, (b) Norepinephrine via β_3-adrenergic receptor → cAMP → PKA → lipase activation
- When activated, allows H^+ to flow back from IMS to matrix bypassing F_0F_1-ATPase
- Result: Proton gradient → heat (not ATP)
- Simultaneously, FFA released from TAG serve as fuel for β-oxidation → sustains ETC and H^+ pumping

Clinical importance: Newborns (especially premature) depend on BAT for thermoregulation — cannot shiver. Around neck/scapular region.

What is the difference between Phase I and Phase II reactions in xenobiotic metabolism? PYQ 2016-S, 2023

Phase I (Functionalisation): Introduces or exposes a functional group (-OH, $-NH_2$, -COOH) on the xenobiotic. Reactions: oxidation (CYP450 monooxygenase — most common), reduction, hydrolysis. Makes molecule slightly more polar. May produce reactive/toxic intermediates. Location: smooth ER of liver.

Phase II (Conjugation/Biosynthetic): Adds large polar group to the Phase I product. Reactions:

- Glucuronidation (UDP-glucuronate — most common in humans)
- Sulphation (PAPS — 3'-phosphoadenosine-5'-phosphosulphate)
- Glutathione conjugation (GSH — protects against reactive intermediates)
- Acetylation (acetyl-CoA — by N-acetyltransferase, e.g., isoniazid)
- Amino acid conjugation (glycine for bile acids)
- Methylation (SAM donor)

Phase II always makes compound MORE water-soluble → excretion in urine or bile. "Conjugation reactions are generally detoxification reactions."

Why is the malate-aspartate shuttle more efficient than the glycerophosphate shuttle? `PYQ 2014`

Both shuttles transfer cytoplasmic NADH electrons into the mitochondria (since IMM is impermeable to NADH).

Malate-aspartate shuttle: Cytoplasmic NADH → converted to mitochondrial NADH (equivalent) → enters ETC at Complex I → generates **2.5 ATP per NADH**.

Glycerophosphate shuttle: Cytoplasmic NADH → converted to $FADH_2$ (on IMM by mitochondrial G3P dehydrogenase) → enters ETC at CoQ (bypasses Complex I) → generates only **1.5 ATP per NADH**.

Difference: Malate-aspartate delivers electrons to NAD^+ in matrix (enters via Complex I = 2.5 ATP). G3P shuttle delivers to FAD (enters via CoQ = 1.5 ATP). Complex I proton-pumping step is bypassed in G3P shuttle → fewer ATP made.

Malate-aspartate: heart muscle, liver (high metabolic demand). G3P shuttle: brain, skeletal muscle (faster but less efficient).

13 Essay Question Templates ● TIER 1

📝 EQ — CHEMIOSMOTIC THEORY EXPLAINS MECHANISM OF OXIDATIVE PHOSPHORYLATION [PYQ 2023]

The chemiosmotic theory explains the mechanism of oxidative phosphorylation. PYQ 2023

1. Proposed by: Peter Mitchell (1961) — Nobel Prize in Chemistry (1978). Unified explanation for how electron transport is coupled to ATP synthesis.
2. Core Concept: Energy from electron transport is not stored as a phosphorylated intermediate (as was previously thought) but as an **electrochemical proton gradient** across the inner mitochondrial membrane.
3. Proton Pumping: Complexes I, III, and IV use energy from electron transfer to pump H^+ from the matrix into the intermembrane space. This creates: (a) Chemical gradient: ΔpH (IMS is acidic, matrix is alkaline); (b) Electrical gradient: ΔΨ (IMS positive, matrix negative). Together these constitute the **Proton Motive Force (PMF ≈ 200 mV)**.
4. ATP Synthesis: H^+ can only re-enter the matrix through F_0F_1-ATP synthase. The flow of H^+ through F_0 rotates the c-ring → γ-subunit rotation in F_1 → conformational changes in three β-subunits (Open→Loose→Tight) → ADP + Pi → ATP released. Approximately 3 H^+ per ATP synthesised.
5. Key Requirement: Intact, impermeable IMM is essential. If membrane disrupted (e.g., by detergents) or protons bypass F_0 (e.g., via uncouplers like DNP/Thermogenin) → PMF dissipated as heat → no ATP synthesis.
6. Experimental Evidence: (a) Mitoplasts make ATP when artificial pH gradient applied without any substrates; (b) Reconstituted liposomes with bacteriorhodopsin + ATP synthase make ATP on illumination; (c) Oligomycin blocks F_0 H^+ channel → ATP synthesis stops and ETC slows due to backpressure.

📝 EQ — 2,4-DINITROPHENOL FUNCTIONS AS UNCOUPLER OF RESPIRATORY CHAIN [PYQ 2022]

2,4-Dinitrophenol functions as an uncoupler of the respiratory chain. `PYQ 2022`

1. Definition of Uncoupler: A compound that allows electron transport (and O_2 consumption) to continue but prevents the proton gradient from driving ATP synthesis — the free energy is released as heat.
2. Chemical Nature of DNP: 2,4-Dinitrophenol is a lipophilic weak acid (pKa ~4.0). At physiological pH, it can exist in protonated form (neutral, lipophilic — crosses IMM) and deprotonated form (charged, cannot cross).
3. Mechanism: In the IMS (acidic, high H^+): DNP^- picks up H^+ → becomes HDNP (neutral, lipophilic) → diffuses through IMM into matrix → releases H^+ (becomes DNP^- again in alkaline matrix) → cycles back. Net effect: H^+ transferred from IMS to matrix bypassing F_0F_1 → gradient dissipated as heat.
4. Consequences: ↑ O_2 consumption (ETC uninhibited), ↑ heat production (hyperthermia), ↓ ATP synthesis (ADP accumulates → stimulates ETC further), ↑ substrate oxidation (TCA, β-oxidation stimulated).
5. Historical Use and Toxicity: Used as weight-loss drug in 1930s — weight loss because fats and carbohydrates oxidised but ATP not made → stored energy wasted as heat. BANNED due to fatalities from hyperthermia, cataracts, peripheral neuropathy. Narrow toxic range.
6. Physiological Analogue: Thermogenin/UCP-1 in brown adipose tissue performs the same function under controlled, physiologically beneficial circumstances (non-shivering thermogenesis in newborns and cold-adapted individuals).

📝 EQ — BROWN ADIPOSE TISSUE PROMOTES THERMOGENESIS [PYQ 2011, 2016-S, 2022]

Brown adipose tissue promotes thermogenesis. `PYQ 2011, 2016-S, 2022`

1. Structural Basis: BAT is brown due to abundant mitochondria (iron-rich cytochromes), multilocular lipid droplets, dense capillarity, and high expression of UCP-1 (thermogenin) in the IMM. White adipose has one large lipid droplet, few mitochondria.
2. Mechanism (UCP-1 pathway): Cold exposure → sympathetic NS → norepinephrine → β_3-adrenergic receptor → Gs protein → adenylate cyclase → ↑ cAMP → PKA → phosphorylates/activates hormone-sensitive lipase → TAG hydrolysis → free fatty acids released.
3. UCP-1 Activation: Free fatty acids directly activate UCP-1 (an uncoupling protein that forms an H^+ channel in IMM). H^+ flows from IMS → matrix through UCP-1, bypassing F_0F_1. Proton gradient dissipated → ATP NOT made → energy released as HEAT.
4. Simultaneous Fuel Supply: The same FFA released by lipolysis enter β-oxidation → acetyl-CoA → TCA → generates $NADH/FADH_2$ → feeds ETC → maintains H^+ pumping → sustains thermogenesis.
5. Physiological Importance: Non-shivering thermogenesis in newborns (cannot shiver — underdeveloped muscle). BAT located around neck, interscapular region, kidneys. Premature infants lack sufficient BAT → hypothermia. Also active in hibernating mammals, cold-adapted adults.
6. Therapeutic Potential: Activating BAT (via β_3-agonists, cold exposure) increases energy expenditure → potential anti-obesity strategy. Active area of research — BAT persists in adults around clavicle and mediastinum.

💬 🔥 EXAM DAY HIGH-YIELD QUICKFIRE — MODULE 07

Which complex does NOT pump protons?

Complex II (Succinate Dehydrogenase) — also enters at CoQ not Complex I

Cyanide mechanism

Binds Fe^{3+} of Cytochrome a_3 (Complex IV) → blocks electron transfer to O_2

Oligomycin target

F_0 subunit of ATP synthase — blocks H^+ channel → indirectly slows ETC

Rotenone target

Complex I — blocks Fe-S → CoQ electron transfer

Antimycin A target

Complex III — blocks Q cycle (Qi site)

Malate shuttle ATP yield

2.5 ATP per cytoplasmic NADH (delivers NADH to matrix)

G3P shuttle ATP yield

1.5 ATP per cytoplasmic NADH (delivers $FADH_2$, bypasses Complex I)

Thermogenin location

Inner mitochondrial membrane of brown adipose tissue

DNP mechanism

Lipophilic protonophore — carries H^+ across IMM → dissipates gradient as heat

Most common Phase II reaction

Glucuronidation (UDP-glucuronate donor)

CYP450 overall reaction

$RH + O_2 + NADPH \rightarrow R\text{-}OH + H_2O + NADP^+$ (monooxygenase)

Superoxide dismutase converts...

$O_2{\bullet}^- \rightarrow H_2O_2$ (then Catalase/GPx → H_2O)

Most reactive ROS

Hydroxyl radical (•OH) — generated by Fenton reaction ($Fe^{2+} + H_2O_2$)

Cyt c dual role

ETC: electron carrier (Complex III → IV). Apoptosis: activates apoptosome → caspase cascade

P/O ratio for NADH vs $FADH_2$

NADH = 2.5 ATP; $FADH_2$ = 1.5 ATP (modern values)

MODULE 07 MASTER MNEMONICS

ETC INHIBITORS BY COMPLEX

"Rotenone Murders Antimycin's Cyanide Oligomycin" — I, II, III, IV, V

H^+ PUMPING COMPLEXES

"1, 3, 4 pump protons — Complex 2 does ZERO" (I pump 4, III pump 4, IV pump 2)

SHUTTLES

"Malate = More ATP (heart). Glycerophosphate = Less ATP (brain accepts less)"

PHASE II REACTIONS

"GGSMA" — Glucuronidation, Glutathione, Sulphation, Methylation, Acetylation/Amino acid conjugation

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MODULE 07 · BIOLOGICAL OXIDATION

Caffeine & Cadaver · MBBS Biochemistry Notes · 2010-2025

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